

1 Introduction

In guitar, tone control is a major technical skill. Each musical instrument’s quality of sound—its timbre—is differentiated by the varying composition of harmonic frequencies ($f_n = nf_1$ for $n = 1, 2, 3\dots$), which are present in addition to the fundamental frequency f_1 . In addition to the fundamental variation between instruments, the timbre also varies based on how notes are played (Perov, Polievkt et al., 2016). It is qualitatively observed that plucking near the bridge will create a harsher, “tinny” sound while plucking toward the bridge will create a fuller sound.

This investigation will explore how the quality of sound may change depending on where the string is plucked. The research question I have decided to investigate is: *How does changing the position of the plectrum along a guitar scale affect the frequency of the harmonic with the highest intensity when a string with constant tension is plucked?*

1.1 Background

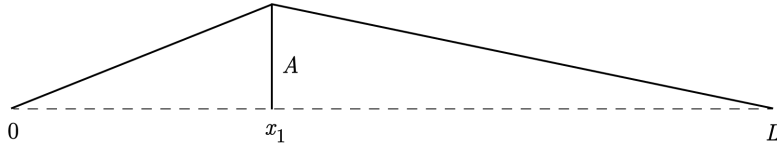


Figure 1: A guitar string of length L pulled a distance A at position x_1

A guitar string is a string with two fixed ends. When the string is plucked (as in Figure 1), a transverse wave is initiated that propagates along the length of the string. The reflection of the transverse wave at the fixed ends superimposes with the original wave, forming nodes (points of zero displacement) and antinodes (points of maximum displacement). This creates a standing wave, which is a wave that does not propagate but rather oscillates in place.

The first harmonic is a standing wave with a single loop made of two nodes and one antinode, which results in a sound wave at the fundamental frequency, given by the equation

$$f_1 = \frac{v}{2L}.$$

The linear principle of superposition states that the net displacement of a system subjected to multiple independent disturbances (waves, in this case) is the algebraic sum of the displacements due to each disturbance. If $y_1(x, t)$ and $y_2(x, t)$ are two waveforms propagating along the same string, then the net waveform $y(x, t)$ is given by equation

$$y(x, t) = y_1(x, t) + y_2(x, t) \quad (1)$$

Due to this principle, multiple, independent standing waves are formed on the guitar string, which results in higher-order harmonics. Each subsequent harmonic is an integer multiple of the fundamental frequency, given by $f_n = \frac{v}{\lambda_n}$ where $\lambda_n = \frac{2L}{n}$. This is known as frequency quantization.

These individual waveforms superimpose linearly to form a composite waveform which is the sum of all individual harmonic waveforms. As a result, the sound emitted by a guitar string is a mixture of multiple harmonic frequencies.

To investigate how the picking length affects the harmonic with the highest amplitude, we must determine the variation of the amplitude of each harmonic frequency separately. The amplitude b_n of the n th harmonic is given by the equation

$$b_n = \frac{2A \sin(n\pi \frac{x_1}{L})}{\frac{x_1}{L}(1 - \frac{x_1}{L})\pi^2 n^2} \quad (2)$$

(Perov, Polievkt et al., 2016)

1.2 Hypothesis

Calculating b_n (defined in Equation 2) for each $n = 0...20$ in the background, it is hypothesized that

2 Methodology

2.1 Variables

	Area	Parameters
test	$\pi h \frac{D^2 - d^2}{4} \quad (3)$	h : height D : outer radius d : inner radius
test	$\frac{\sqrt{2}}{12} a^3 \quad (4)$	a : edge length

Table 1: Variables

2.1.1 Pilot Study

2.2 Apparatus & Experimental Setup

- 6 stringed electric guitar (customized 6 string *Kiesel Guitars “Osiris”* model used)
- 1 set of 0.009 – 0.042mm gauge nylon electric guitar strings (*D’Addario XS Nickel “XSE0942 Super Light Coated Electric Guitar Strings”* model used)
- 1 audio interface with instrument input (*Audient EVO4* used)
- 1 instrument cable
- Computer with audio analysis software (*MATLAB* used)
- Tape measure
- Clamp stand

Figure 2 presents the experimental setup used to collect data. The guitar is clamped down for ease of plucking, and the tape measure is placed directly on the 2nd and 3rd strings to be on the same axis as the 1st string (which is plucked).



Figure 2: Experimental setup

2.3 Experimental Procedure

2.4 Safety, Ethical, and Environmental Considerations

3 Analysis

3.1 Raw Data and Observations

3.1.1 Raw Data

3.1.2 Observations

3.1.3 Fourier Analysis

3.2 Processed Data

3.3 Calculations and Uncertainty

3.4 Graphical Analysis

4 Conclusion

4.1 Scientific Reasoning

5 Evaluation

5.1 Strengths

5.2 Limitations

5.3 Further Study

References

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